

Mathematical Documentation - Zoning Plan Generator Algorithm

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1. Mathematical Foundations

1.1 Notation and Definitions

Sets and Spaces

- $\mathbf{B} = \{b, b, \dots, b\}$: Set of all blocks, where $n = |\mathbf{B}|$
- $\mathbf{S} = \{s, s, \dots, s\}$: Set of all superblocks, where $m = |\mathbf{S}|$
- $\mathbf{F} = \{f, f, \dots, f\}$: Set of building features (cores, anchors)
- $\mathbf{G}$: Grid space 2 with dimensions $W \times H$ (width $\times$ height)
- $\mathbf{C} = \{\text{ME, WE, US, SUPPORT, SPECIALTY}\}$: Set of block categories

Block Properties For each block $b \in \mathbf{B}$: - **ID** : $b.\text{id}$ (unique identifier) - **Category** : $b.\text{cat} \in \mathbf{C}$ - **Dimensions** : $(w, h) \in \mathbb{R}^2$ (width, height in meters) - **Area** : $a = w \times h$ - **Position** : $p = (x, y) \in \mathbf{G}$ - **Superblock Assignment** : $s \in \mathbf{S}$ - **Placement Status** : $\text{placed}(b) \in \{\text{true, false}\}$

Building Features For each feature $f \in \mathbf{F}$: - **Type** : $f.\text{type} \in \{\text{core, anchor, special}\}$ - **Position** : $p = (x, y) \in \mathbf{G}$ - **Dimensions** : $(w, h) \in \mathbb{R}^2$ - **Footprint** : $F = \{(x, y) \mid x \leq x + w, y \leq y + h\}$

Adjacency Rules

- $\mathbf{R} = \{r, r, \dots, r\}$: Set of adjacency rules
 - Each rule $r = (p, \text{from}, \text{to}, \text{rel}, w)$ where:
 - $p \in \{0, 1, 2, 3, 4, 5\}$: Priority level
 - from, to : Block types or features
 - $\text{rel} \in \{\text{must_avoid, must_be_near, preferred_near, should_avoid}\}$
 - $w \in [-2, 1]$: Weight/score modifier
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2. Problem Formulation

2.1 Formal Problem Statement

Given: - Set of blocks B with properties (category, dimensions) - Grid space G with dimensions $W \times H$ - Set of building features F with fixed positions - Set of adjacency rules R with priorities - Target superblock compositions $T = \{t, t, \dots, t\}$

Find: - Position assignment $P : B \rightarrow G$ - Superblock assignment $\Sigma : B \rightarrow S$

Subject to:

1. Non-overlap Constraint:

$$b_i, b_j : B_i \cap B_j = \emptyset$$

where B_i is the footprint of block b_i

2. Grid Boundary Constraint:

$$b : 0 \leq x < W, 0 \leq y < H$$

3. Feature Avoidance Constraint:

$$b \cap f : B \cap F = \emptyset$$

4. Priority 0 (Non-compromisable) Constraints:

$$r : (r, B, F)$$

where r is the constraint satisfaction function and θ is the threshold

Optimize:

Maximize the objective function:

$$Z = \alpha \cdot \Phi_{\text{geometric}} + \beta \cdot \Phi_{\text{adjacency}} + \gamma \cdot \Phi_{\text{composition}} + \delta \cdot \Phi_{\text{placement}}$$

where $\alpha, \beta, \gamma, \delta$ are weighting coefficients with $\sum = 1$

2.2 Constraint Hierarchy

Priority levels form a strict hierarchy:

$$P_1 > P_2 > P_3 > P_4 > P_5 > P_6$$

Where: - P_1 : Critical constraints (must satisfy) - P_2 - P_6 : Soft constraints (maximize satisfaction)

3. Constraint Functions

3.1 Geometric Rectangularity Constraint (Priority 1)

Aspect Ratio Function:

$$AR(b) = \max(w, h) / \min(w, h)$$

Rectangularity Score:

$$R(b) = \begin{cases} 1, & \text{if } AR(b) \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Overall Geometric Score:

$$\Phi_{\text{geometric}} = (\sum R(b)) / |B_{\text{placed}}| \times 100$$

$$\text{where } B_{\text{placed}} = \{b \in B \mid \text{placed}(b) = \text{true}\}$$

3.2 Distance Functions

Euclidean Distance:

$$d(p, p') = \sqrt{(x_p - x_{p'})^2 + (y_p - y_{p'})^2}$$

Manhattan Distance:

$$d_M(p, p') = |x_p - x_{p'}| + |y_p - y_{p'}|$$

Centroid Distance: For blocks with dimensions:

$$c = (x + w/2, y + h/2)$$

$$d_c(b, b') = d(c, c')$$

3.3 Adjacency Satisfaction Function

For a rule $r = (p, \text{from}, \text{to}, \text{rel}, w)$:

Distance-based Satisfaction:

$$(r, b, f) = \begin{cases} w \times k, & \text{if } \text{rel} = \text{"must_avoid"} \quad d(p, p') \geq _avoid \\ w \times k, & \text{if } \text{rel} = \text{"must_be_near"} \quad d(p, p') < _near \\ -k, & \text{if } \text{rel} = \text{"must_avoid"} \quad d(p, p') < _avoid \\ -k, & \text{if } \text{rel} = \text{"must_be_near"} \quad d(p, p') \geq _far \\ w \times k, & \text{if } \text{rel} = \text{"preferred_near"} \quad d(p, p') < _pref \\ w \times f(d(p, p')), & \text{otherwise} \end{cases}$$

where: $_avoid = 10\text{m}$ (avoidance distance threshold) - $_near = 15\text{m}$ (near distance threshold) - $_far = 30\text{m}$ (far distance threshold) - $_pref = 20\text{m}$ (pre-

ferred distance threshold) - k are penalty/reward multipliers - $f(d)$ is a continuous decay function

Decay Function:

$$f(d) = e^{(-d/\lambda)}$$

where λ is the decay constant (default: 10)

3.4 Overall Adjacency Score

For a single block:

$$A(b) = \sum_r R[(r, b, \text{target}(r))]$$

For all placed blocks:

$$\Phi_{\text{adjacency}} = (\sum A(b)) / |B_{\text{placed}}|$$

Normalized to percentage:

$$\Phi_{\text{adjacency_norm}} = \text{sigmoid}(\Phi_{\text{adjacency}}) \times 100$$

where $\text{sigmoid}(x) = 1/(1 + e^{-x})$

4. Optimization Objectives

4.1 Superblock Composition Matching

Target Composition: For superblock s , target composition:

$$t = \{t_{\text{ME}}, t_{\text{WE}}, t_{\text{US}}, t_{\text{SUPPORT}}, t_{\text{SPECIALTY}}\}$$

where $\sum_c C t_c = 1$

Actual Composition:

$$a_{c} = |\{b \in s \mid b.\text{cat} = c\}| / |s|$$

Composition Error:

$$E(s) = \sum_c C |t_c - a_c|$$

Composition Match Score:

$$M(s) = \max(0, 1 - E(s)) \times 100$$

Overall Composition Score:

$$\Phi_{\text{composition}} = (\sum M(s)) / |S|$$

4.2 Placement Efficiency

Placement Rate:

$$\Phi_{\text{placement}} = |B_{\text{placed}}| / |B| \times 100$$

Area Utilization:

$$U_{\text{area}} = (\sum B_{\text{placed}} a) / (W \times H) \times 100$$

Compactness Metric: For superblock s with blocks $\{b_1, b_2, \dots, b_n\}$:

$$\text{Compactness}(s) = \text{Area}(\text{ConvexHull}(s)) / \sum_{b \in s} \text{Area}(b)$$

where $\text{ConvexHull}(s)$ is the convex hull of all blocks in s

5. Scoring Functions

5.1 Priority-based Scoring

Priority Weight Function:

$$(p) = \begin{cases} 10, & \text{if } p = 0 \text{ (non-compromisable)} \\ 5, & \text{if } p = 1 \text{ (geometric)} \\ 3, & \text{if } p = 2 \text{ (block-to-feature)} \\ 2, & \text{if } p = 3 \text{ (block-to-block)} \\ 1, & \text{if } p = 4 \text{ (superblock-to-superblock)} \\ 0.5, & \text{if } p = 5 \text{ (block-to-department)} \end{cases}$$

Weighted Adjacency Score:

$$A_{\text{weighted}}(b) = \sum_r R[(r.p) \times (r, b, \text{target}(r))]$$

5.2 Placement Evaluation Function

For candidate position (x, y) for block b :

$$\begin{aligned} \text{Score}(b, x, y) = & \cdot S_{\text{adjacency}}(b, x, y) \\ & + \cdot S_{\text{geometry}}(b, x, y) \\ & + \cdot S_{\text{proximity}}(b, x, y) \\ & + \cdot S_{\text{superblock}}(b, x, y) \end{aligned}$$

where: - $S_{\text{adjacency}}$: Adjacency rule satisfaction - S_{geometry} : Geometric quality (rectangularity, alignment) - $S_{\text{proximity}}$: Distance to similar blocks - $S_{\text{superblock}}$: Cohesion with superblock members - $\cdot$: Weighting coefficients

Adjacency Component:

$$S_{\text{adjacency}}(b, x, y) = \sum_r R[(r, (x, y), \text{target}(r))]$$

Geometry Component:

$$S_{\text{geometry}}(b, x, y) = \alpha \cdot \text{Alignment}(x, y) + \beta \cdot \text{Rectangularity}(b)$$

where Alignment measures grid alignment:

$$\text{Alignment}(x, y) = \begin{cases} 1, & \text{if } (x \bmod \text{grid_unit} = 0) \text{ and } (y \bmod \text{grid_unit} = 0) \\ 0.5, & \text{if } (x \bmod \text{grid_unit} = 0) \text{ or } (y \bmod \text{grid_unit} = 0) \\ 0, & \text{otherwise} \end{cases}$$

Proximity Component:

$$S_{\text{proximity}}(b, x, y) = -\sum_b B_{\text{same_cat}} [1/d((x,y), p)]$$

where $B_{\text{same_cat}}$ are blocks of same category

Superblock Component:

$$S_{\text{superblock}}(b, x, y) = -\text{mean}(\{d((x,y), p) \mid b \in s(b)\})$$

where $s(b)$ is the superblock containing b

5.3 Threshold Functions**Acceptance Threshold:**

$$_accept(b) = \begin{cases} -1, & \text{if } b.\text{cat} \in \{\text{ME}, \text{WE}\} \text{ (stricter for critical blocks)} \\ -3, & \text{if } b.\text{cat} \in \{\text{US}\} \\ -5, & \text{if } b.\text{cat} \in \{\text{SUPPORT}, \text{SPECIALTY}\} \end{cases}$$

Placement Decision:

$$\text{Place}(b, x, y) = \begin{cases} \text{true}, & \text{if } \text{Score}(b, x, y) \geq _accept(b) \text{ and } \text{Valid}(b, x, y) \\ \text{false}, & \text{otherwise} \end{cases}$$

where Valid checks non-overlap and boundary constraints

6. Statistical Measures**6.1 Variance and Standard Deviation****Block Size Variance:**

$$\sigma_{\text{size}}^2 = (1/n) \sum (a_i - \bar{a})^2$$

where $\bar{a} = (1/n) \sum a$

Position Distribution:

$$\begin{aligned} \sigma_x^2 &= (1/n) \sum (x - \bar{x})^2 \\ \sigma_y^2 &= (1/n) \sum (y - \bar{y})^2 \end{aligned}$$

6.2 Category Distribution

Chi-Square Test for Uniformity:

$$\chi^2 = \sum_c C [(O_c - E_c)^2 / E_c]$$

where: O_c = observed count of category c - E_c = expected count ($|B|/|C|$)

Entropy of Distribution:

$$H(B) = -\sum_c C [p_c \times \log(p_c)]$$

where $p_c = |\{b \in B \mid b.cat = c\}| / |B|$

6.3 Correlation Measures

Category Co-location:

$$\text{Corr}(c, c) = \text{Cov}(X_c, X_c) / (\sigma_c \times \sigma_c)$$

where X_c is the position vector for category c

6.4 Performance Metrics

Precision (Block Placement):

$$\text{Precision} = |B_{\text{placed}} \cap B_{\text{correct}}| / |B_{\text{placed}}|$$

Recall:

$$\text{Recall} = |B_{\text{placed}} \cap B_{\text{correct}}| / |B|$$

F1-Score:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

7. Geometric Calculations

7.1 Collision Detection

AABB (Axis-Aligned Bounding Box) Intersection:

For blocks b at (x, y) and b at (x, y) :

$$\text{Overlap}(b, b) = (x < x + w) \quad (x + w > x) \\ (y < y + h) \quad (y + h > y)$$

Separation Vector:

```

sep_x = min(|x + w - x|, |x + w - x|)
sep_y = min(|y + h - y|, |y + h - y|)
sep = min(sep_x, sep_y)

```

7.2 Area Calculations**Block Coverage:**

$$\text{Coverage}(\mathbf{B_placed}, G) = (\sum \mathbf{B_placed} \ a) / \text{Area}(G)$$
Void Space:

$$\text{Void} = \text{Area}(G) - \sum \mathbf{B_placed} \ a - \sum f \ \text{Area}(f)$$
Fragmentation Index:

$$\text{FI} = \text{Number_of_disconnected_regions} / |\mathbf{B_placed}|$$
7.3 Convex Hull**Jarvis March Algorithm for Convex Hull:**

For set of points $P = \{p, p, \dots, p\}$:

1. Find leftmost point p
2. Current point $h = p$
3. Repeat:
 - Find point q such that all other points are counter-clockwise to (h, q)
 - Add q to hull
 - $h = q$
4. Until $h = p$

Convex Hull Area (Shoelace Formula):

$$\text{Area} = (1/2) |\sum (x_i y_{i+1} - x_{i+1} y_i)|$$
7.4 Packing Density

Local Density: For block b in neighborhood radius r :

$$_local(b, r) = \sum_b N(b, r) \ a / (\pi \times r^2)$$

where $N(b, r) = \{b \mid d(p, p) \leq r\}$

Global Density:

$$_global = |\mathbf{B_placed}| / (W \times H)$$

8. Complexity Analysis

8.1 Time Complexity

Data Generation: - Block generation: $O(n)$ - Superblock creation: $O(n \times m)$ where m is number of superblocks

Placement Algorithm: - Worst case: $O(n \times W \times H \times |R|)$ - Average case: $O(n \times k \times |R|)$ where k is average attempts per block - Best case: $\Omega(n)$ if all blocks place immediately

Adjacency Evaluation: - Per block: $O(|R| \times |F|)$ - Total: $O(n \times |R| \times |F|)$

Analysis: - Geometric rectangularity: $O(n)$ - Adjacency compliance: $O(n \times |R| \times |F|)$ - Composition matching: $O(m \times n)$ - Total analysis: $O(n \times (m + |R| \times |F|))$

8.2 Space Complexity

Grid Storage: - Grid: $O(W \times H)$ - Blocks: $O(n)$ - Features: $O(|F|)$ - Rules: $O(|R|)$ - Total: $O(W \times H + n + |F| + |R|)$

9. Convergence Properties

9.1 Greedy Placement Guarantee

The algorithm uses a greedy heuristic with backtracking:

Theorem: If a valid placement exists, the algorithm will find A placement (not necessarily optimal) with probability approaching 1 as the number of restart attempts increases.

Proof Sketch: 1. Each placement attempt is independent 2. Valid placements form a non-empty subset of position space 3. Random restart explores different regions 4. Probability of finding valid placement $P(\text{success}) > 0$ 5. After k attempts: $P(\text{success}) = 1 - (1 - p)^k \rightarrow 1$ as $k \rightarrow \infty$

9.2 Optimality Bounds

Lower Bound on Placement Rate:

$$\Phi_{\text{placement}} = (W \times H - \sum \text{Area}(f)) / (\sum a) \times 100$$

when no adjacency constraints exist

Upper Bound on Composition Match:

$$M(s) = 100 - E_{\text{min}} \times 100$$

where E_{min} is minimum achievable error given discrete block counts

10. Stochastic Properties

10.1 Random Variables

Block Dimensions:

$W \sim \text{Uniform}(w_{\min}, w_{\max})$

$H \sim \text{Uniform}(h_{\min}, h_{\max})$

Category Selection:

$\text{Cat} \sim \text{Categorical}(p_{\text{ME}}, p_{\text{WE}}, p_{\text{US}}, p_{\text{SUPPORT}}, p_{\text{SPECIALTY}})$

where $\sum p = 1$

10.2 Expected Values

Expected Block Area:

$E[A] = E[W] \times E[H] = ((w_{\min} + w_{\max})/2) \times ((h_{\min} + h_{\max})/2)$

Expected Placement Success:

$E[\text{Success}] = \int \int P(\text{valid}(x,y) \mid \text{constraints}) dx dy$

10.3 Monte Carlo Estimation

Empirical Placement Rate: After N trials:

$\Phi_{\text{placement}} = (1/N) \sum (|B_{\text{placed}}^{(i)}| / |B|) \times 100$

Confidence Interval:

$\text{CI} = \Phi_{\text{placement}} \pm z_{(\alpha/2)} \times (\hat{\sigma}/\sqrt{N})$

where $z_{(\alpha/2)}$ is critical value and $\hat{\sigma}$ is sample standard deviation

11. Mathematical Properties

11.1 Monotonicity

Score Monotonicity: If $d < d'$ for must_avoid constraint:

$\text{Score}(d) < \text{Score}(d')$

Composition Error Monotonicity:

$E(\text{composition})$ is monotonically decreasing with better matching

11.2 Boundedness

All scores are bounded:

$-\infty < A(b) < \infty$ (but typically in $[-20, 20]$)
 $0 \leq M(s) \leq 100$
 $0 \leq \Phi_x \leq 100$ for all metrics Φ

11.3 Continuity

Distance-based scores are continuous:

$\lim_{d \rightarrow d'} f(d) = f(d')$

Adjacency score is piecewise continuous: Discontinuities only at threshold boundaries

12. Numerical Methods

12.1 Discretization

Grid Discretization: Continuous space $(0, W) \times (0, H)$ discretized to integer lattice

Resolution:

$\Delta x = 1$ meter
 $\Delta y = 1$ meter

12.2 Optimization Techniques

Gradient-free Optimization: - Greedy heuristic with scoring - Local search with random restarts - No explicit gradient computation

Search Strategy:

1. Start at superblock anchor position
2. Scan in raster order (left-to-right, top-to-bottom)
3. Evaluate score at each valid position
4. Accept first position above threshold
5. If no valid position, try relaxed threshold

13. Validation Metrics

13.1 Constraint Satisfaction Rate

For priority level p :

$CSR(p) = |\{r \in R_p \mid \text{satisfied}(r)\}| / |R_p| \times 100$

13.2 Quality Metrics

Overall Quality Score:

$$Q = w \cdot \text{CSR}(0) + w \cdot \text{CSR}(1) + w \cdot \text{CSR}(2) + \dots + w \cdot \text{CSR}(5) \\ + w_g \cdot \Phi_{\text{geometric}} + w_c \cdot \Phi_{\text{composition}} + w_p \cdot \Phi_{\text{placement}}$$

where $\sum w = 1$

13.3 Robustness Measures

Sensitivity Analysis:

$$S_{\text{param}} = Q / \text{param}$$

for each parameter

Stability:

$\text{Var}(Q)$ across multiple runs with same input

Appendix A: Symbol Reference

Symbol	Description	Domain
B	Set of blocks	Finite set
S	Set of superblocks	Finite set
F	Set of features	Finite set
R	Set of adjacency rules	Finite set
G	Grid space	$\mathbb{Z}^2$
W, H	Grid dimensions	
w, h	Block dimensions	
a	Block area	
p	Position	$\mathbb{Z}^2 \setminus \{ \}$
d(p, q)	Distance function	
	Constraint function	
	Satisfaction function	
Φ	Objective function	[0, 100]
	Threshold	
w, w_g, w_c, w_p	Weights	[0, 1]

Appendix B: Constants and Parameters

Parameter	Default Value	Range	Description
_avoid	10 m	[5, 20]	Avoidance distance
_near	15 m	[10, 25]	Near distance
_far	30 m	[20, 50]	Far distance
_pref	20 m	[15, 30]	Preferred distance
	10	[5, 20]	Decay constant
k	2	[1, 5]	Avoidance reward
k	3	[2, 5]	Proximity reward
k	10	[5, 20]	Avoidance penalty
k	5	[2, 10]	Distance penalty
k	2	[1, 3]	Preference reward
grid_unit	1 m	[0.5, 2]	Grid alignment
AR_threshold	2.0	[1.5, 3.0]	Aspect ratio limit

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References: See Algorithm Documentation for implementation details